Modeling Daily Soil Moisture in Lethbridge, AB: A Water Balance Approach

UNIVERSITY OF GUELPH

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Geography, Environment & Geomatics

BACKGROUND

Soil moisture regulates land-climate interactions by controlling evapotranspiration, energy cycles, and water availability for vegetation. Shifts in soil moisture under warming conditions have led to notable changes in land cover and growing capabilities[1].

In Canada, the Prairie ecoregion provides important ecosystem services and contributes substantially to the national economy through agricultural production[2]. Altered soil moisture dynamics pose a threat to the many benefits provided by this region and need to be further investigated to understand how best to mitigate the effects[3]. Direct measurement of soil properties can be costly and resource intensive; however, model-based approaches offer an alternative by using existing weather station data to represent hydrological processes[4].

The model presented here uses meteorological data from the Lethbridge, AB weather station[5] to simulate daily soil moisture over a 12-year period (2014 – 2026).

APPROACH

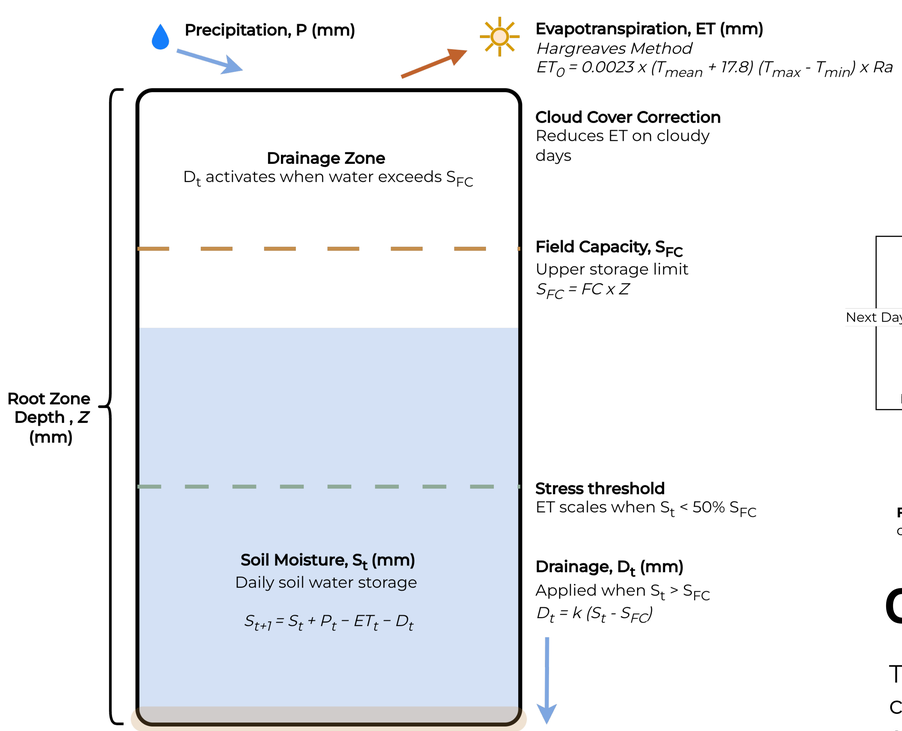

Figure 1. Conceptual diagram of the daily water balance bucket model based on Rodriguez-Iturbe et al.[6] Precipitation enters the root zone storage, which is depleted by evapotranspiration, estimated using the Hargreaves method and adjusted for cloud cover. Drainage is applied when storage exceeds field capacity. The stress threshold represents the minimum storage level before ET_0 is scaled down to reflect reduced plant water availability.

ASSUMPTIONS

- **Soil properties are uniform and static** – field capacity, drainage coefficient, and root zone depth are constant
- **Precipitation falls as rain –** snowmelt is not modeled
- **Root zone depth is fixed –** seasonal variation is not represented
- **Cloud cover correction is uniform** – the same reduction factor applies to all cloudy periods
- **No lateral flow or surface runoff** – all water is either lost to evapotranspiration, drains vertically, or remains in storage

MODELING PROCESS

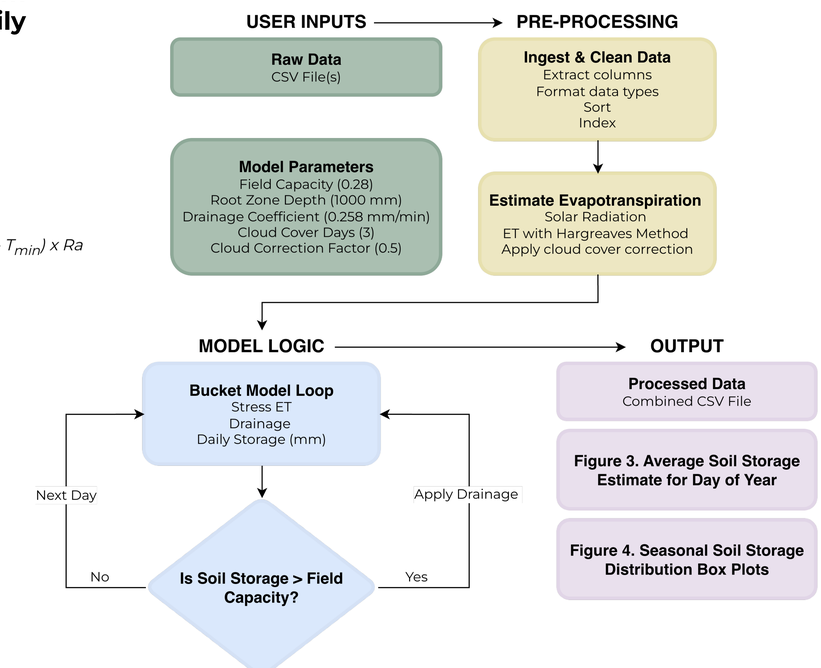

Figure 2. Workflow diagram of the daily soil moisture model. All major steps including data ingestion, calculations and the main model logic used to simulate daily water storage are shown.

OUTPUT

The model produces a CSV file containing daily calculations of solar radiation, evapotranspiration, drainage estimates, and total soil moisture for the period covered by the input data. The model also reports the average soil moisture value for each day of the year (Figure 3), and the seasonal distribution of soil moisture values (Figure 4).

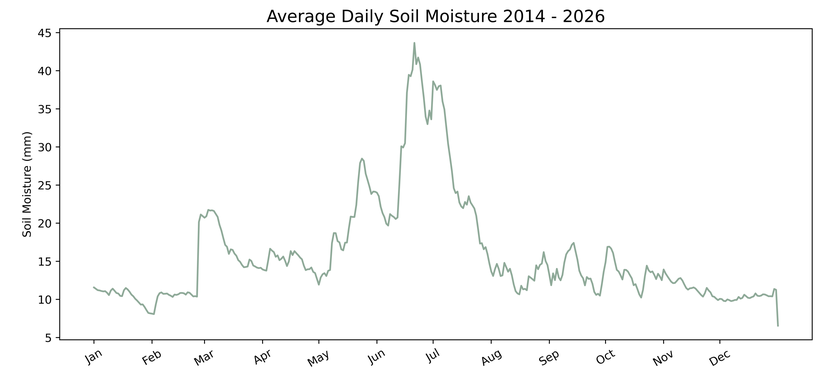

Figure 3. Average daily soil moisture (mm) by day of year (2014 – 2026). Values represent mean soil moisture for each calendar day, averaged across all years, revealing seasonal patterns.

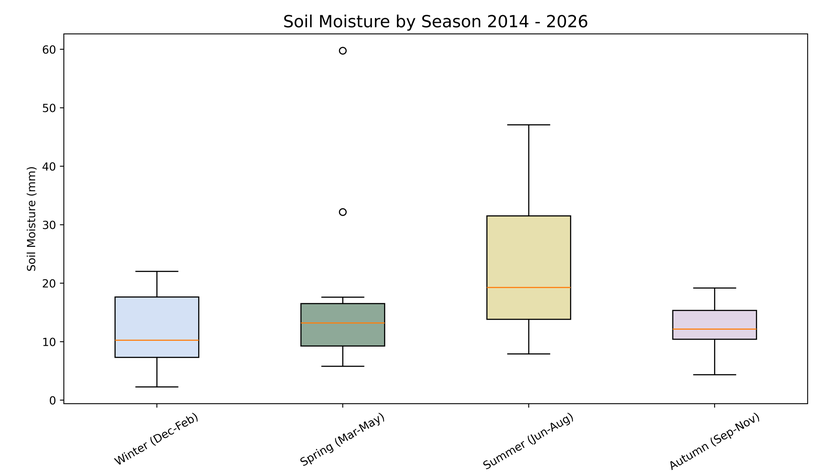

Figure 4. Seasonal distribution of mean soil moisture (mm) (2014 – 2026). Box plots show the distribution of annual seasonal averages. Each value corresponds to the mean soil moisture for a given season within a year. Seasonal differences and interannual variability are reflected in the spread and outliers.

STRENGTHS & LIMITATIONS

The model provides an interpretable approach to simulating soil moisture using meteorological data while capturing key water balance processes. The soil moisture stress function improves realism by reducing evapotranspiration under low water availability, and the model reproduces reasonable seasonal patterns. However, simplifying assumptions, such as uniform soil properties, fixed root zone depth, and a constant cloud correction, limit accuracy. The exclusion of snow dynamics and lateral flow or runoff further constrains realism.

REFERENCES

1. Seneviratne et al. (2010). Earth-Science Reviews
2. Chipanshi et al. (2021). Intl. Journal of Climatology
3. Hanesiak et al. (2011). Atmosphere-Ocean
4. CORDIS. (2020). Impact of soil moisture on climate change projections
5. Government of Alberta. (2026). Acis.alberta.ca
6. Rodriguez-Iturbe et al. (1999). Proc. R. Soc. Lond. A

ACKNOWLEDGEMENTS

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